

The DUNE FD-VD light simulation for low-energy physics: workflow, five simulation bugs, and proposed fixes

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Bugs B, C, D, E and N1 were found by Xin Qian (fdvd_sim studies, WireCell/wcp-porting-validation); this note reproduces them, fixes them in `duneopdet`, and measures the effect of each fix.

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Abstract

The photon-detector (light) simulation of the DUNE far detector has five bugs in two `duneopdet` modules, `SIPMOpSensorSim` and `WaveformDigitizerSim`. Two of them only matter when several interactions share one readout, which is exactly the situation of low-energy (solar, supernova) physics, where a neutrino sits on top of the radiological background. In FD-VD solar events simulated with the official configuration and background, 12.38 % of the OpHits are copies of other OpHits, and the neutrino’s argon light is 0.655 of what a per-track late-light reference gives; neither effect exists in signal-only samples, so a calibration made on signal-only samples is biased for real conditions. This note (1) explains in plain language what the low-energy light workflow is and how it differs from the beam workflow, (2) gives a minimum `gen` $\rightarrow$ `Geant4` $\rightarrow$ `detsim` workflow that reproduces every bug in about an hour, and (3) for each bug shows the effect before and after a proposed fix on 20 signal-only and 20 signal-plus-background events. The fixes are one patch to `duneopdet` (it applies to `v10_26_00d00` and to `develop`); every fix can be switched off to reproduce existing samples bit for bit.

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1 Summary for the conveners

Table 1 lists the bugs. The letters follow Xin Qian’s numbering (fdvd_sim doc 30), so they can be cross-referenced with his notes. “Mixed” means a MARLEY solar- ν_e event plus the nominal DUNE radiological model in the same readout; “solar-only” means the MARLEY event alone.

Table 1: The five bugs, their size in our samples before (*legacy*) and after (*fixed*) the fix, and the proposed default. Numbers are totals over the 20 mixed and 20 solar-only events of section 4.

	module	what goes wrong	size, before $\rightarrow$ after	default
B	WaveformDigitizerSim	overlapping “focus ranges”: the same waveform samples are stored twice and get the noise twice	dup. OpHits 12.38 % $\rightarrow$ 0.00 % (mixed)	fix on
C	SIPMOpSensorSim	“late light” is timed from the first photon of the whole event, not of the interaction	MARLEY Ar PE/photon 0.138 $\rightarrow$ 0.210 (mixed)	opt-in
D	SIPMOpSensorSim	dark-count times are in μs but stored as ns	100.0 % $\rightarrow$ 0.2 % of dark counts within $\pm 5 \mu\text{s}$ of the ν	fix on
E	WaveformDigitizerSim	unsigned underflow of the readout start at the beginning of the window	188 $\rightarrow$ 0 snippets at $2.95 \times 10^{17} \mu\text{s}$	always
N1	SIPMOpSensorSim	photon detection is a second Poisson draw instead of a binomial one	$P(0 \text{ PE} 1 \text{ photon})$ 0.689 $\rightarrow$ 0.627	fix on

- **B and C are pile-up bugs.** They need several interactions in one readout and are absent in signal-only samples. They change the light of the *signal* in mixed events, not only the background.
- **D, E and N1 act in every event.** D moves the dark counts onto the neutrino flash. E creates a handful of waveform snippets with a nonsense time stamp (and reads memory outside the waveform buffer). N1 makes the photon counting statistics too wide.
- **Proposal.** One commit on branch `fix-light-sim-pileup` of github.com/HaiwangYu/duneopdet, on top of DUNE/duneopdet develop (appendix A). B, D and N1 are fixed by default with a switch back to the legacy behaviour; E has no switch (the legacy output is undefined behaviour); C changes the light model and is opt-in (`LateLightReference: "Track"`), to be decided by the photon-detector simulation experts.
- **Other known issues** of the same chain, not fixed here, are listed in section 5. The largest light loss in the chain is not a bug but a setting of the OpHit finder (F in that list).

2 Low-energy versus beam physics, in plain words

Beam neutrinos. The accelerator sends a short spill of neutrinos every $\sim 1 \text{ s}$, and we know exactly when. The detector is read out around the spill time. A beam neutrino interaction deposits GeV of energy, produces long tracks and showers, and gives a big flash of scintillation light. Radioactive decays in the argon also happen during the readout, but they deposit only MeV each, far below the beam event, so they barely matter. In simulation, one event is usually one neutrino interaction at $t = 0$.

Low-energy neutrinos (solar, supernova). A solar ν_e interacting by charged current on argon ($\nu_e + {}^{40}\text{Ar} \rightarrow e^- + {}^{40}\text{K}^*$, simulated with MARLEY) deposits 5–20 MeV: a short electron track plus a few de-excitation γ s. There is no external clock: the neutrino can arrive at any time, and the detector must trigger on itself. And the signal is *not* much bigger than the background. The argon itself is radioactive (${}^{39}\text{Ar}$, ${}^{85}\text{Kr}$, ${}^{42}\text{Ar}$, radon chains), the detector materials and the cavern emit γ s and neutrons: the DUNE radiological model puts about 130 000 decays into one 4.25 ms readout of the FD-VD workspace. Figure 1 shows the light of one simulated readout: the MARLEY flash (205 PE) is one of many flashes in a sea of 1,179,934 background PE.

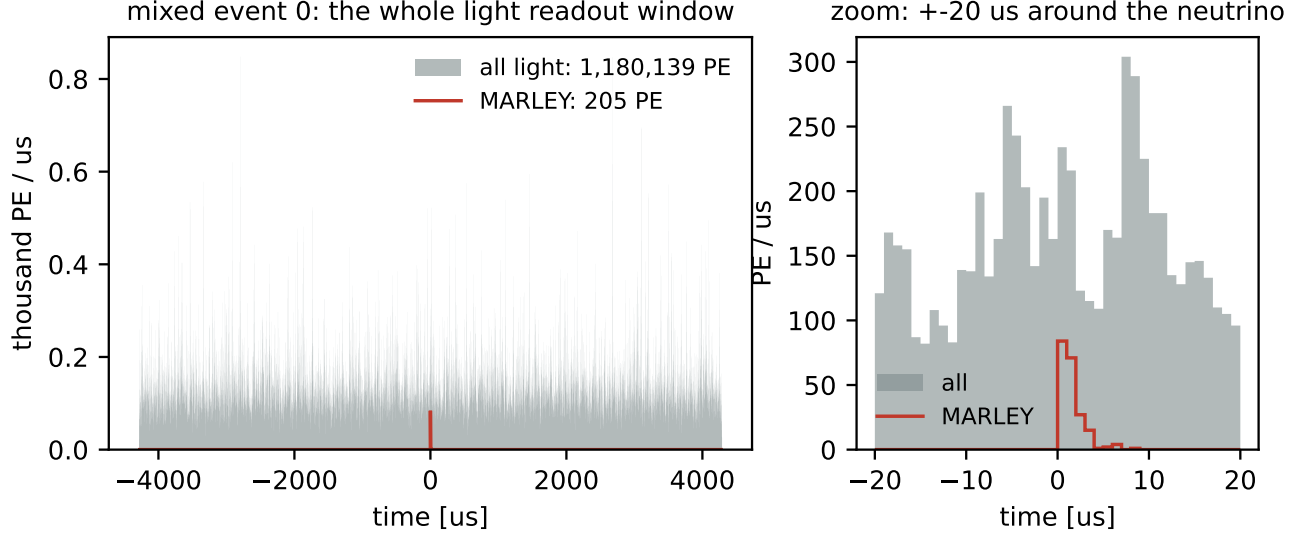


Figure 1: Photoelectrons (PE) detected in one mixed event (stock simulation), per μs . Grey: all light; red: light from the MARLEY neutrino at $t = 0$. Left: the whole ± 4.25 ms light readout. Right: zoom on the neutrino.

What this means for the simulation.

- **The background must be simulated in the same event as the signal** (“mixed” events). Signal and background share photon detectors and electronics: pulses overlap, triggers merge, noise is shared. Simulating them separately and adding the outputs is not the same thing.
- **The readout is long:** 8500 TPC samples of $0.5 \mu\text{s} = 4.25$ ms for the charge, ± 4.25 ms for the light. The light window is wider because the charge of a neutrino can arrive up to one drift time (4 ms) after its light.
- **The light matters more.** The flash gives the neutrino time, which is needed to know how far the charge drifted, i.e. its position and its energy (charge lost to impurities on the way). With ~ 1000 flashes per readout, picking the right flash (charge–light matching) is the central problem of the low-energy reconstruction.
- **Code paths that never see pile-up in beam samples see it constantly here.** Bugs B and C are of this kind: invisible in signal-only samples, large in mixed ones.

Table 2: Beam versus low-energy workflow.

	beam	low energy (solar, supernova)
signal energy	0.5–10 GeV	5–30 MeV
time of the interaction	known (beam spill)	unknown; self-trigger on light or charge
background per readout	negligible next to the signal	$\sim 130\,000$ radioactive decays, 1,171,788 PE of light against 490 PE from a solar ν
simulation unit	one interaction	signal + background in one readout (“mixed”)
generator	GENIE	MARLEY + DUNE radiological model (Decay0/RadioGen)
role of the light	t_0 , trigger	t_0 , energy, and picking 1 flash out of ~ 1000
reconstruction (official)	standard reco1/reco2	solar_reco: gaushit, ophit10ppm, SolarOpFlash, SolarCluster, SolarEvent, analysis in SolarNuAna

The official low-energy reconstruction (S. Manthey Corchado, DUNE solar group) lives in the packages `dunereco/`, `LowEUtils`, `duneopdet/LowEPDSUtils` and `duneana/SolarNuAna`. Its job files are `solar_reco_dunevd10kt_1x8x14_3view_30deg.fcl` (FD-VD) and `solar_reco_dune10kt_1x2x6.fcl` (FD-HD). This note stops at the OpHits, which are the input of that reconstruction.

3 The light simulation workflow

3.1 The detector

The geometry is the FD-VD workspace `dunevd10kt_1x8x14_3view_30deg`: one drift volume of the vertical-drift module (2.56kt of argon, 6.5m drift), read out by $8 \times 14 = 112$ charge-readout modules at the top and **184 X-ARAPUCA photon detectors**: 112 on the cathode, 56 on the long side walls and 16 on the short ones. The argon is doped with 10 ppm of xenon, which shifts part of the slow argon light to xenon light. Every photon detector (“OpDet”) is read by one electronics channel (“OpChannel”).

3.2 The chain, step by step

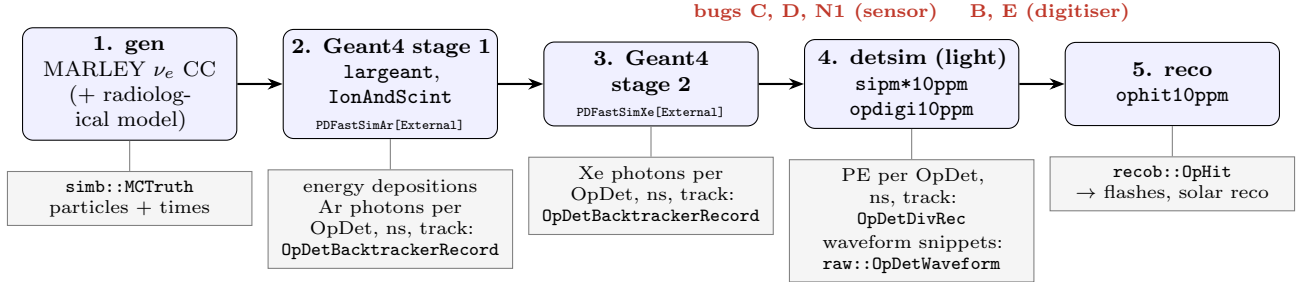


Figure 2: The FD-VD light simulation chain (dunesw v10.26.00d00) and the product each step writes. The bugs of this note live in step 4.

1. **Generation.** MARLEY produces the electron and γ s of one solar ν_e CC interaction at $t = 0$, at a random point in the active volume, with a flat 2–30 MeV neutrino energy (physical spectra are obtained by reweighting). For mixed events, about 30 more generator modules add the radiological decays of the DUNE model, spread uniformly over ± 4.25 ms. Job: `prodmarley_solar_cc_flat[_radiological_decay0]_dunevd10kt_1x8x14_3view_30deg.fcl`.
2. **Geant4, stage 1.** Tracks every particle through the argon (`largeant`), converts the energy depositions into ionisation electrons and scintillation photons (`IonAndScint`), drifts the electrons (`elecDrift`) and computes how many *argon* scintillation photons reach each OpDet and when (`PDFastSimAr`; `PDFastSimArExternal` for light made outside the field cage). Photons are not tracked one by one: a semi-analytic model (inside the field cage) or a photon library (outside) gives the fraction of photons from each point that reaches each OpDet. Only 5 % of the physical photons are simulated (`ScintPreScale` 0.05); the sensor simulation corrects for it. The output, `OpDetBacktrackerRecord` (“BTR”), is a list per OpDet of (arrival time in ns, Geant4 track ID, number of photons). One such entry is an “SDP”.
3. **Geant4, stage 2.** The same for the xenon light (`PDFastSimXe`, `PDFastSimXeExternal`).
4. **Light detector simulation** (“detsim”). Two kinds of modules:
 - `SIPMOpSensorSim`, run four times (`sipmAr10ppm`, `sipmXe10ppm`, `sipmAr10ppmExt`, `sipmXe10ppmExt`), one per BTR product. It decides which photons become photoelectrons (PE): detection efficiency 0.027 (3 % X-ARAPUCA $\times$ 90 % mesh transmission), a wavelength correction (0.65 for Ar light, 0.69 for Xe light), for Ar light a “late light” factor 0.31 (bug C), SiPM cross-talk (20 % chance that one PE becomes two), and dark counts (10 Hz, bug D). Output: `OpDetDivRec`, the PE per OpDet, time and track.
 - `WaveformDigitizerSim` (`opdigi10ppm`) turns the PE of all four products into one waveform per channel: 62.5 MHz (16 ns samples), a 10 ADC pulse per PE that decays in 0.2 μ s, pedestal 500, line noise 2.6 ADC RMS, 14 bits. To save time, it only simulates the samples around PE pulses (“focus ranges”, bug B). It then mimics the DAPHNE self-trigger: when the waveform rises by more than 1.5 PE within 10 samples, it stores a 320-sample snippet starting 20 samples before the trigger (bug E). Output: `raw::OpDetWaveform`, the snippets.
5. **OpHit finding.** `ophit10ppm` (`OpHitFinder` with larana’s `AlgoSiPM`) finds pulses above 15 ADC that stay above threshold for at least 60 samples, and converts their area into PE (area/130 + 0.43). Output: `recob::OpHit`. Flash finding and the solar reconstruction start from here.

3.3 Running it: the minimum workflow

All code is in `FD-LE/00-op-sim-debug/work` (github.com/HaiwangYu/FD-LE). `dunesw v10` is built for Scientific Linux 7; on the AlmaLinux 9 `dunegpvm/dunebuild` machines every LArSoft command runs inside the Fermilab SL7 container, which `env/in-sl7.sh` starts and sets up. One event through the whole chain:

```
cd FD-LE/00-op-sim-debug/work
scripts/run_chain.sh sol 0 # solar-only event 0: ~2.5 min, 1 GB
scripts/run_chain.sh mix 0 # mixed event 0: ~10 min, 6.2 GB peak (Geant4)
```

Each call runs the five steps above with the official job files (light-only versions of `detsim` and `reco` in `fcl/`) and writes `data/events/<sol|mix>/evt<i>/reco.root`. Seeds are fixed per event and step (`masterSeed = 100000/200000 + 10 × index + step`, Xin Qian’s scheme), so event `mix 0` here is the same event as in his samples: we reproduce his numbers for it to the last digit (section 6).

To compare simulation versions on identical photons, the light simulation is *rerun* from the BTRs kept in `reco.root`, with the event’s own `detsim` seed:

```
scripts/build_duneopdet.sh # duneopdet v10_26_00d00 + patches/duneopdet-light-sim-fixes.patch (~2 min)
scripts/run_rerun.sh stock mix 0 # CVMFS build
scripts/run_rerun.sh fixB mix 0 # patched build, only the fix for bug B switched on
scripts/run_all.sh # everything in this note: 20+20 events x 8 configurations + plots (~1.5 h)
```

To look inside an art file (from `env/in-sl7.sh bash`): `lar -c eventdump.fcl -s reco.root -n 1` lists the data products, `config_dumper -P reco.root` prints the configuration each job ran with, and the `gallery` Python bindings used in `analysis/dump_light.py` read products without writing an art job.

The configurations are listed in table 3. Each rerun ends with `analysis/dump_light.py`, which writes the numbers per event (`light.npz`); `analysis/make_plots.py` makes every figure and number of this note (`numbers.tex`).

Table 3: Rerun configurations. “legacy” is the patched build with every switch at its old value; it differs from the stock build only by the fix for bug E, which has no switch. Each `fix*` configuration switches on one fix.

configuration	build	settings
stock	CVMFS <code>duneopdet v10_26_00d00</code>	–
legacy	patched	all switches legacy (E fixed)
fixB	patched	legacy + <code>MergeOverlappingRanges: true</code>
fixC	patched	legacy + <code>LateLightReference: "Track"</code> (Ar products)
fixD	patched	legacy + <code>DarkNoiseTimeInNs: true</code>
fixN1	patched	legacy + <code>BinomialQE: true</code>
default	patched	the proposed defaults: B, D, E, N1 fixed, C legacy
all	patched	default + <code>LateLightReference: "Track"</code>

4 The bugs, their impact, and the fixes

The samples: solar-only events 0–19 and mixed events 0–19 (20 + 20). Per event the stock simulation gives 490 PE (solar-only, all MARLEY) and 1,171,788 PE (mixed), 221,591 snippets and 12,929 OpHits. “Before” is the `legacy` configuration and “after” the matching `fix*` configuration, except for bug E (stock versus legacy).

4.1 Bug B — overlapping focus ranges: samples stored twice, duplicate OpHits, double noise

What goes wrong. For each PE, `WaveformDigitizerSim` marks a “focus range” (the pulse plus 100 samples of margin on each side) in which it adds noise and looks for triggers. `FocusList::AddRange` is meant to merge ranges that overlap, but it misses two cases: a new range that *contains* an old one is added as a second range, and a range that grows until it reaches its neighbour is never merged with it. Everything after that loops over the ranges, so in an overlap the noise is added twice and the trigger fires twice on the same samples: the same piece of waveform is stored in two snippets, and the OpHit finder finds the same pulse twice.

Impact. Overlaps need many PE close in time on one channel, which only the radiological background provides: solar-only events are unaffected, in mixed events 15.66 % of the stored samples are copies, 12.38 % of the OpHits are duplicates and they carry 15.94 % of the OpHit PE. The duplication does not care where the light came from: **signal pulses are duplicated too**. The doubled noise (median baseline RMS 2.74 instead of 2.50 ADC) fires extra triggers: removing the duplicate hits after the fact leaves 226,600 hits and 5,605,068 PE, still more than the 218,038 hits and 5,348,590 PE of the fixed simulation.

Fix. A new `FocusList.h: AddRange` merges the new range with every range it overlaps (one pass; the ranges are disjoint before each call). If no range ever overlapped, the result, including the order of the ranges, is exactly the legacy one, so events without overlaps are bit-identical. Switch `MergeOverlappingRanges` (default `true`; `false` = legacy). A unit test (`FocusList_test`, 2000 random pile-up trials) checks both properties.

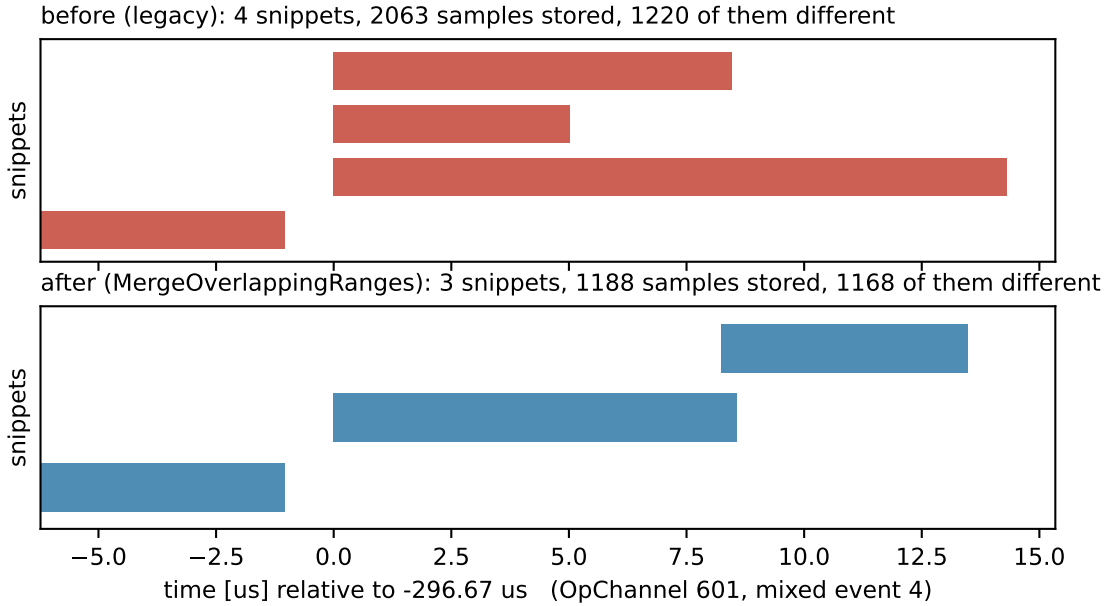


Figure 3: Bug B on one channel of a mixed event: the stored snippets (one bar each) before and after the fix. Before, three snippets start on the same tick and store the same samples again. After, every sample is stored once, except the 20-sample pre-trigger of a snippet that opens right after the previous one closed (N6 in section 5). This place was chosen because both versions read out the same samples there; elsewhere the readouts also differ because the doubled noise of the legacy version fires extra triggers.

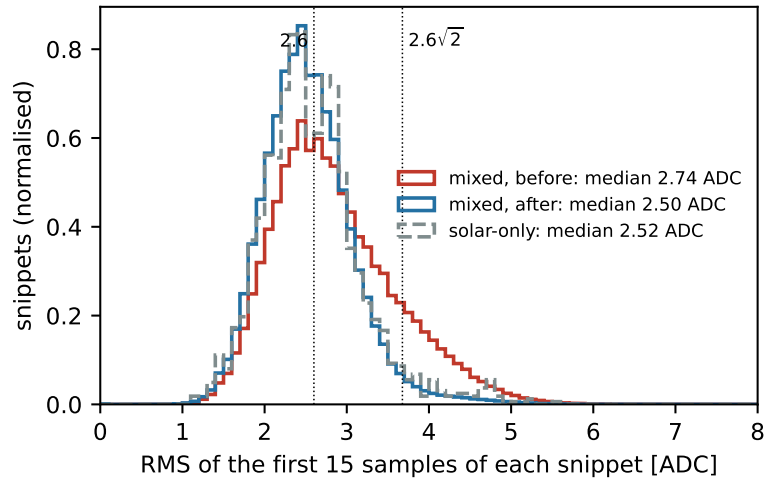
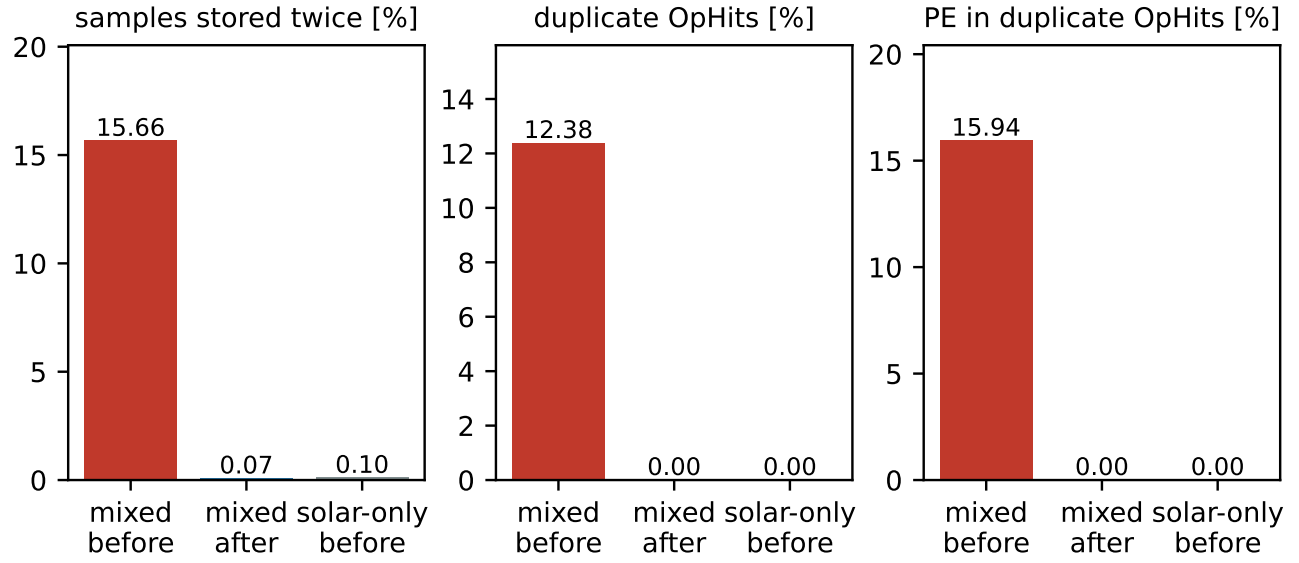


Figure 4: Bug B, all events. Top: fraction of samples stored twice, of duplicate OpHits (same channel and peak time) and of the OpHit PE they carry, before and after the fix. The 0.07 % of samples still stored twice after the fix are the 20-sample pre-trigger of a new snippet re-reading the end of the previous one, which is how the readout is modelled. Bottom: baseline noise of the snippets; the noise is added twice where ranges overlap (expected RMS $2.6\sqrt{2} = 3.7$ ADC).

4.2 Bug C — the “late light” clock starts at the first photon of the whole event

What goes wrong. In xenon-doped argon most of the slow argon light is converted into xenon light, so only a part of it arrives as argon light. `SIPMOpSensorSim` models this with the arrival time: argon photons that arrive within 10 ns of a reference time keep their weight, later ones are scaled by 0.31. The model only makes sense if the reference is the start of the *same* interaction’s light, but the code takes the first photon on that OpDet in the *whole event* (± 4.25 ms). In a mixed event that is a background photon, usually milliseconds before the neutrino, so all argon light of the neutrino, its prompt light included, is treated as late. (A loop meant to find the first time does nothing; it is harmless because the list is already time-ordered.)

Impact. Solar-only events are unaffected (the first photon is the neutrino’s own). In mixed events MARLEY gets 0.138 argon PE per argon photon instead of 0.188. The per-track reference (the fix) raises the MARLEY light (argon and xenon) by 5.0 % in solar-only and by 10.6 % in mixed events, and the total light of mixed events by 20.6 %: the background is dimmed the same way.

Fix. Switch `LateLightReference`: “Event” (default, legacy) or “Track”: the reference is the first photon of the same Geant4 track on that OpDet. Tracks of one interaction start within ns of each other; a delayed process (neutron capture, decay) gets its own reference. Per track is slightly more generous than per interaction (each track of a MARLEY event gets its own 10 ns window); an exact per-interaction reference would need the MC-truth grouping, which the module does not have. We propose “Track” but leave the choice to the light-simulation experts, as it changes the light yield of every sample with xenon doping.

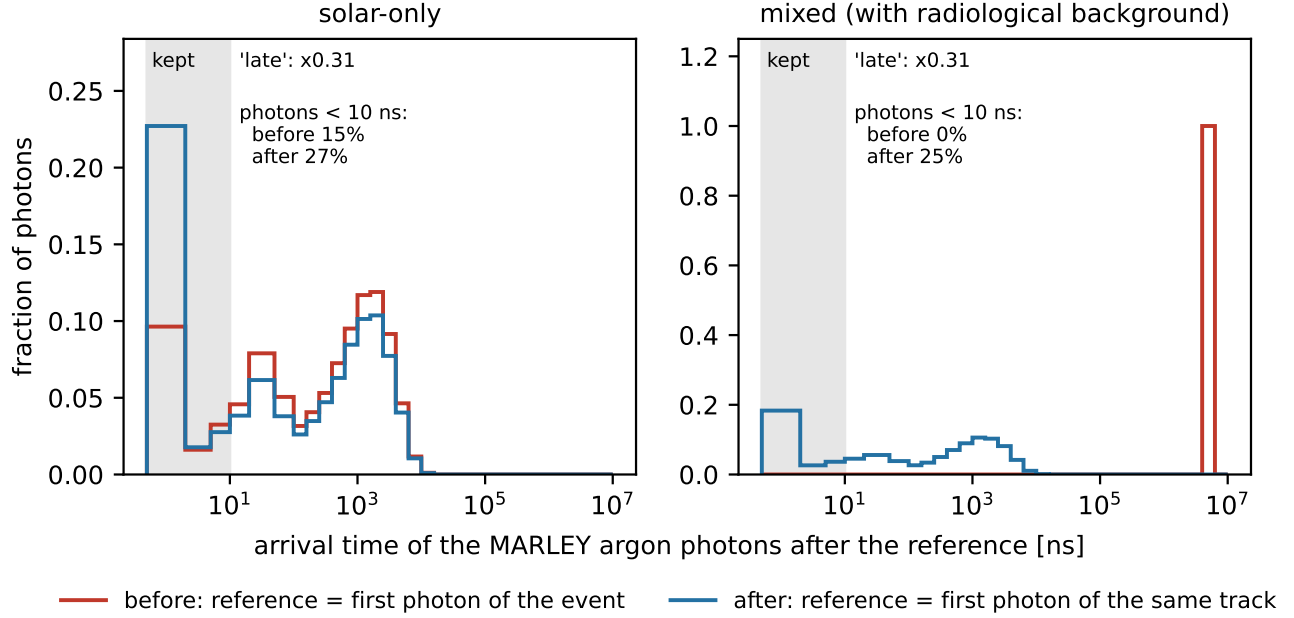


Figure 5: Bug C: arrival time of the MARLEY argon photons after the reference time, with the event’s first photon on the OpDet as reference (legacy, red) and with the first photon of the same track (fix, blue). Photons in the grey band keep their weight, later ones get $\times 0.31$. In mixed events (right) no MARLEY photon is within 10 ns of the event reference.

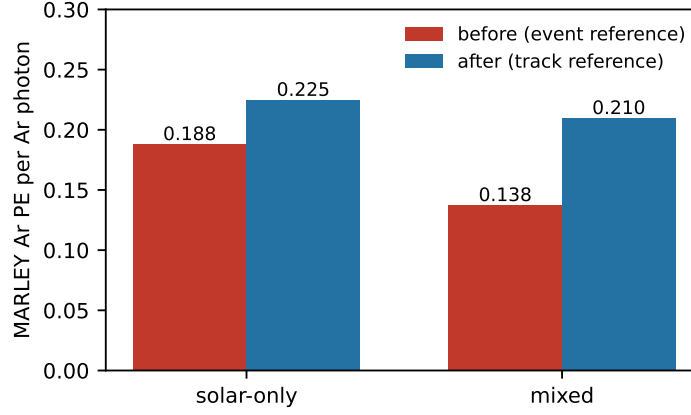


Figure 6: Bug C: MARLEY argon PE per simulated argon photon, before and after the fix. Before, the same neutrino gives 0.655 of its per-track argon light in mixed events; after, the two samples agree within the statistics of 20 events.

4.3 Bug D — dark counts all land on the neutrino

What goes wrong. `SIPMOpSensorSim::AddDarkNoise` draws dark-count times over the readout window, with the window and the mean spacing in μs , and stores them in the `OpDetDivRec`, whose times are in ns. The right number of dark counts is drawn, but squeezed $1000\times$ in time: all of them fall within $\pm 4.25 \mu\text{s}$ of $t = 0$, which is where the neutrino is.

Impact. 38.5 dark PE per event (1,541 in the 20+20 events, from -4.04 to $4.25 \mu\text{s}$) are piled on the signal flash instead of being spread over the -4.04 to $+4.25 \text{ ms}$ range the code intends. They are single PE on many channels, so few make OpHits (Xin Qian measured $\sim 1\%$ on the MARLEY OpHit PE, doc 14), but they bias any study of the flash at the neutrino time, and the effect grows with the dark rate.

Fix. Switch `DarkNoiseTimeInNs` (default `true`): the time is converted to ns before it is stored.

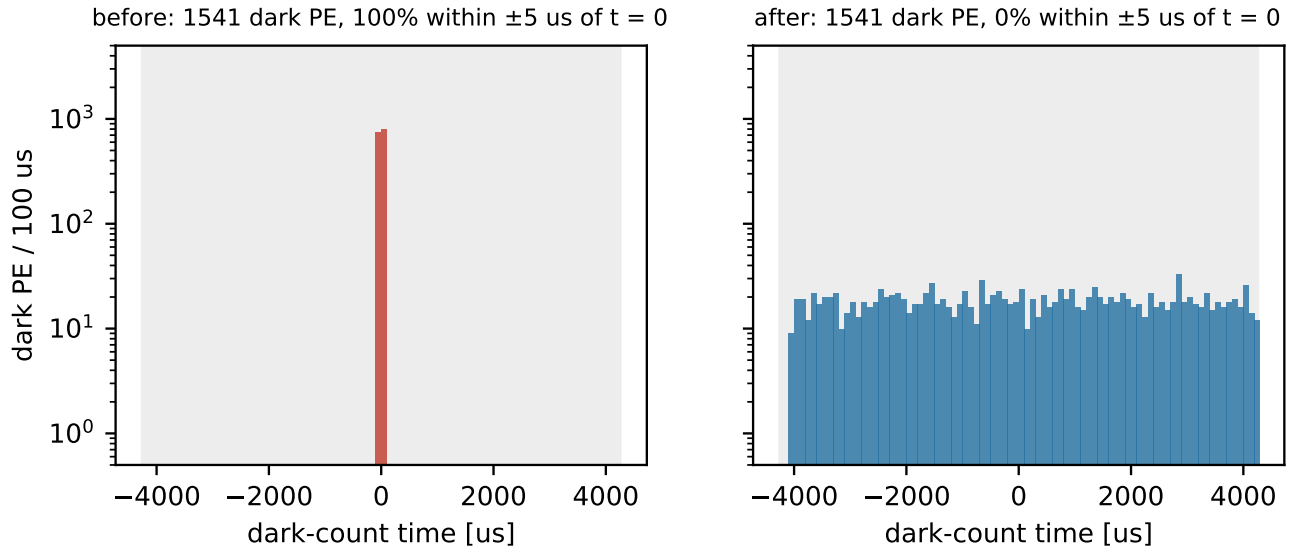


Figure 7: Bug D: times of the dark-count PE, all events, before and after the fix (grey: the light readout window).

4.4 Bug E — readout start underflows at the beginning of the window

What goes wrong. The self-trigger opens a snippet 20 samples before the trigger sample: `wstart = tick - PreTrigger`. Both are unsigned integers, so a trigger in the first 20 samples of the window gives $2^{64} - k$ instead of a negative number. The snippet gets the time stamp 2.95×10^{17} μs , and it is copied from memory before the start of the waveform buffer (an out-of-bounds read, and possibly a write, since the digitiser clamps saturated samples in place).

Impact. Only mixed events have light at the very start of the window: 188 such snippets in the 20 mixed events (69 OpHits at that time), none in solar-only events. Small in physics terms, the hits are easy to drop, but it is undefined behaviour.

Fix. Clamp the start at the first sample. No switch: the legacy output cannot be reproduced reliably anyway. After the fix the same triggers give normal snippets that start at the first sample of the window ($-4255 \mu\text{s}$).

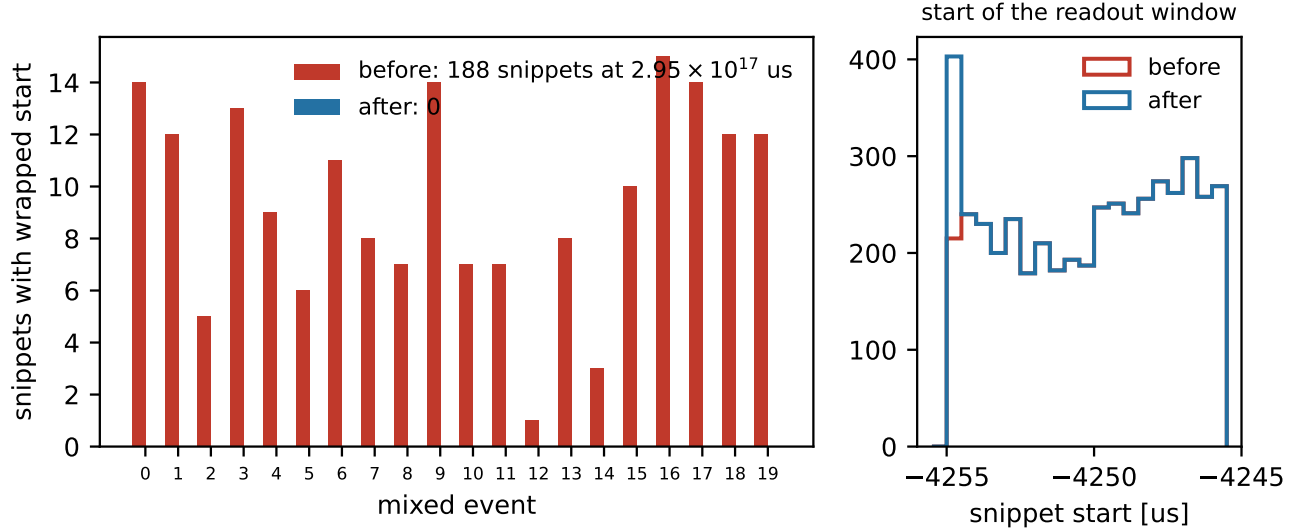


Figure 8: Bug E. Left: snippets with a wrapped start time per mixed event, before and after the fix. Right: snippet start times at the beginning of the window; after the fix the same triggers give snippets that start at $-4255 \mu\text{s}$.

4.5 Bug N1 — photon detection is Poisson on top of Poisson

What goes wrong. PDFastSim already draws the number N of photons that reach an OpDet as a Poisson number. SIPMOpSensorSim then draws the detected photons as $\text{Poisson}(qN)$ (with $q = 0.373$ for xenon light) instead of detecting each of the N photons with probability q (binomial). The mean is right, the spread is not: one photon can give two detected photons, and $P(0 | N = 1) = e^{-q} = 0.689$ instead of $1 - q = 0.627$.

Impact. The PE per OpDet fluctuate more than physically possible (variance $\times (1 + q)/(1 - q)$ before cross-talk). Every mean-based calibration agrees with the legacy simulation, so it hid; it matters for resolutions and for any study of small signals.

Fix. Switch BinomialQE (default true): `CLHEP::RandBinomial(N, q)`. Different random numbers, so not bit-identical to the legacy output when on.

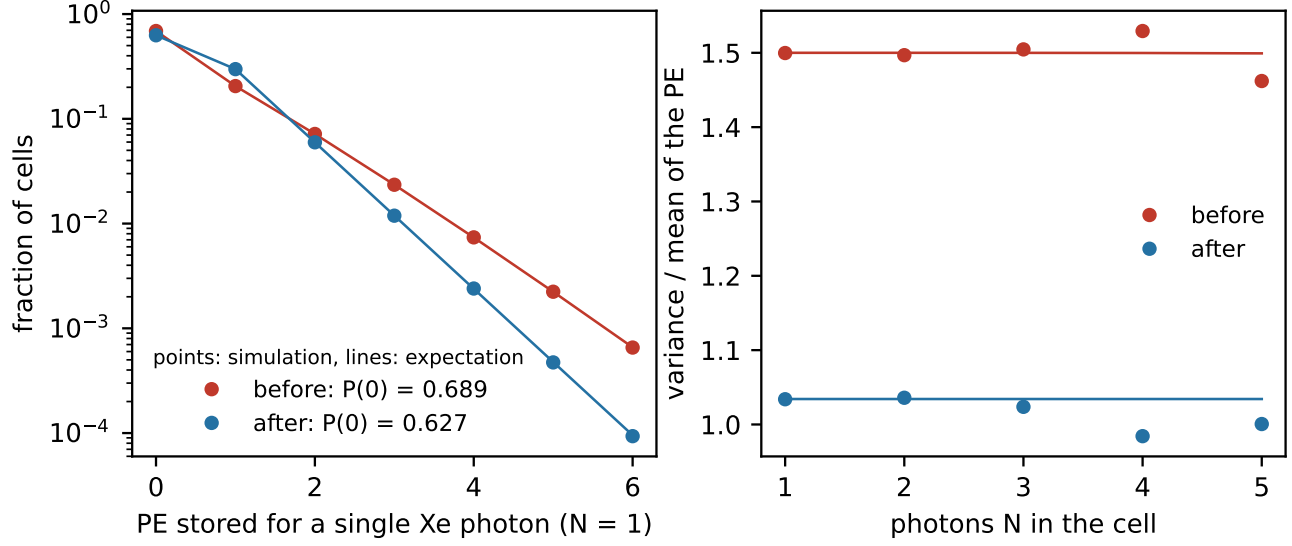


Figure 9: Bug N1, measured on the xenon-light cells that hold exactly one Geant4 entry with N photons (38,231,097 cells with $N = 1$). Left: PE stored for a single photon; lines: the expectation of each model with 20 % cross-talk. Right: variance over mean of the stored PE versus N .

4.6 All fixes together

Table 4: Totals over the samples with the stock build, the proposed defaults, and all fixes including the per-track late light. OpHits with a wrapped time (bug E) are not counted.

	solar-only (20)			mixed (20)		
	stock	default	all	stock	default	all
PE (sensor output)	10,162	10,135	10,627	23,435,770	23,437,995	28,261,229
MARLEY PE	9,791	9,781	10,281	11,765	11,886	13,178
OpHits	181	178	175	258,588	211,736	259,223
OpHit PE	3,320	3,309	3,498	6,667,039	5,295,985	6,473,104

In mixed events the proposed defaults leave the photoelectrons of the sensor simulation unchanged in total (N1 changes only their fluctuations, D only the time of the dark counts) but remove 12.38 % of the OpHits and reduce the OpHit PE from 6,667,039 to 5,295,985: the duplicates of bug B and the extra triggers of its doubled noise. The per-track late light then adds 20.6 % of light. Solar-only events barely change with the defaults; the per-track late light raises their MARLEY light by 5.0 %.

5 Other known issues of the chain (not fixed here)

From Xin Qian's fdvd.sim docs 03, 08 and 30; letters and numbers as there.

- **A** — `OpFlashFinderVerticalDrift` (official `opflash10ppm`) mixes up time-sorted ranks and hit indices when it collects neighbours, so flashes are built from the wrong hits. Reconstruction, not simulation.
- **F** — `ophit10ppm` requires 60 samples ($0.96\mu\text{s}$) above threshold for a pulse that decays in $0.2\mu\text{s}$: most single-PE pulses never make a hit. A setting, and the largest light loss of the chain (Xin Qian: the hits keep 23 % of the PE).
- **N2** — dark noise is added once per sensor module (4 times, 40 Hz in total) and only on OpDets that saw light.
- **N3** — `OpDetDivRec::AddPhoton` (dunecore) credits every photon at an existing time to the first track there; totals are right, per-track attribution is not (negligible in practice).
- **N4** — the residual argon light keeps the $1.6\mu\text{s}$ decay time of undoped argon.

- **N5** — `ophit10ppm` adds 0.43 PE to every `OpHit` (`SPEShift`, tuned for FD-HD).
- **N6** — consecutive snippets of one range can overlap by up to 20 samples (the residual of bug B after the fix).
- The same `FocusList::AddRange` as bug B is copied into five other digitisers, including the ProtoDUNE-HD and ProtoDUNE-VD ones; they would need the same fix and a ProtoDUNE validation.

6 Validation

- **The events are Xin Qian’s events.** With his seeds, our mixed events 0 and 500 have 15 028 912 and 14 910 646 samples stored twice and 31 755 and 31 970 snippets starting on the same tick as another, the numbers of his doc 08a.
- **The rerun reproduces the stored simulation:** stock rerun versus the products of the original job (waveforms and the four `OpDetDivRec` products, compared by hash): 40/40.
- **Each fix changes only what it should.** The legacy configuration differs from the stock build only in the snippets of bug E (wrapped start before, start at the first sample after): 40/40; its sensor output is identical: 40/40. Fix B leaves the sensor output unchanged: 40/40; fix C leaves the xenon products unchanged: 40/40. With fix B on, 17/20 solar-only events are bit-identical to the legacy output, including all 13/20 that have no sample stored twice; the other solar-only events and all mixed events had overlapping ranges.
- **Build:** the patch compiles with `-Werror` against `dunesw v10_26_00d00`, the unit test passes, and the patch applies unchanged to `duneopdet develop` (the two modules have not changed since `v10_26_00d00`).

A The patch

The fixes are commit `6c73f2f` on branch `fix-light-sim-pileup` of the fork `github.com/HaiwangYu/duneopdet`, on top of `DUNE/duneopdet develop` (`497d0a1`, 2026-10-06), ready for a pull request. The same change is the patch `work/patches/duneopdet-light-sim-fixes.patch`, which also applies to `v10_26_00d00` and is what `scripts/build_duneopdet.sh` builds:

- `OpticalDetector/FocusList.h` (new): `FocusList` moved out of `WaveformDigitizerSim_module.cc`, with the merging `AddRange` (B).
- `WaveformDigitizerSim_module.cc`: switch `MergeOverlappingRanges` (B); clamp of the snippet start (E).
- `SIPMOpSensorSim_module.cc`: switches `LateLightReference` (C), `DarkNoiseTimeInNs` (D), `BinomialQE` (N1); the dead loop removed.
- `WaveformDigitizerSim.fcl`, `SIPMOpSensorSim.fcl`: the new switches with their defaults and comments.
- `OpticalDetector/test/FocusList_test.cc` (new): Boost unit test of the range merging.

To reproduce an existing sample bit for bit, set `MergeOverlappingRanges: false`, `DarkNoiseTimeInNs: false` and `BinomialQE: false` (the output then differs only in the bug-E snippets).

B Glossary

art, fcl	the event-processing framework of LArSoft, and its configuration language
BTR, SDP	<code>OpDetBacktrackerRecord</code> : per <code>OpDet</code> , the photons from Geant4 by time and track; one entry is an SDP
DivRec	<code>OpDetDivRec</code> : per <code>OpDet</code> , the photoelectrons by time and track, after the sensor simulation
OpDet, OpChannel	a photon detector (here an X-ARAPUCA), and its electronics channel
PE	photoelectron: one detected photon (a SiPM signal of one unit)
snippet	a stored piece of waveform (<code>raw::OpDetWaveform</code>), 320 samples of 16 ns by default
OpHit, flash	a pulse found in a snippet; a group of <code>OpHits</code> close in time on several <code>OpDets</code>
MARLEY	the generator of MeV neutrino–argon interactions
mixed event	a MARLEY event and the radiological background simulated together in one readout
mrbs, ups	the tools that build (<code>mrbs</code>) and set up (<code>ups</code>) LArSoft packages

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